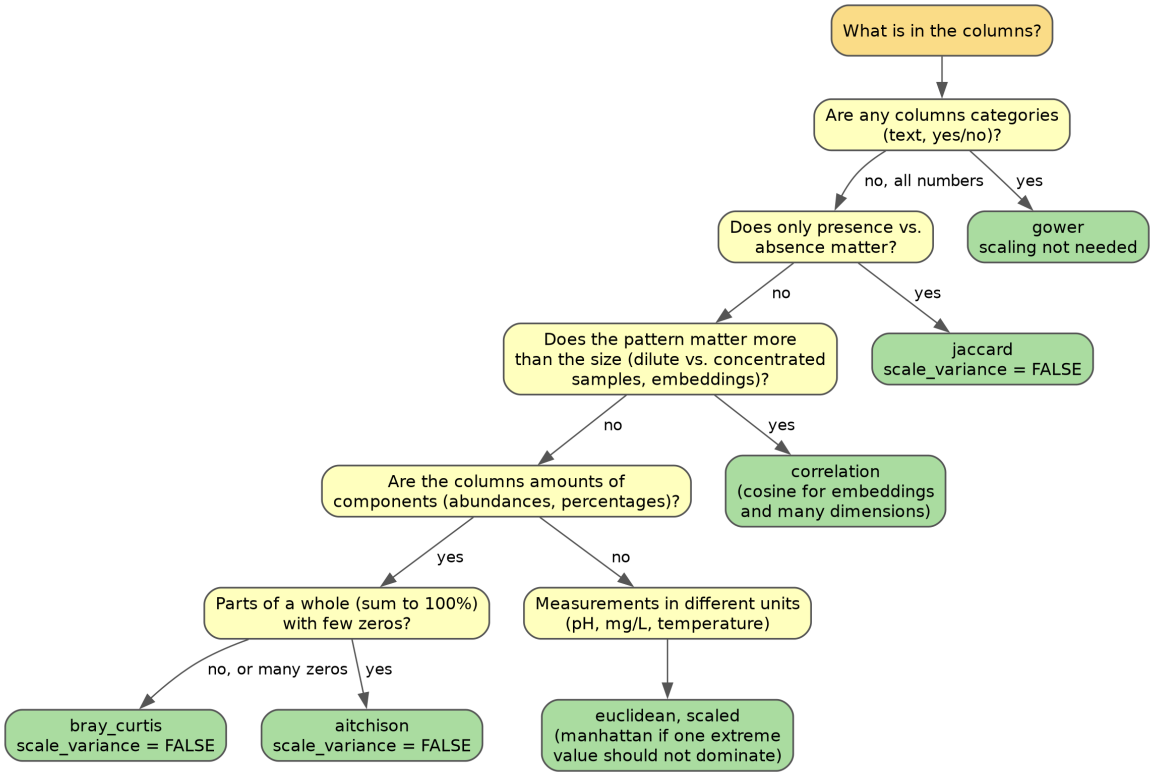
# distances and scaling: summary

Integrated Bioanalytical Chemistry · thebustalab.github.io/integrated_bioanalytics




Two decisions go into every distance matrix. **Scaling decides which variables count**: scale measurements that are in different units so that no single variable dominates, and leave raw amounts unscaled when you want distances to reflect potential differences in the scales that your variables occupy. **The distance metric decides what "similar" means**: the same size and values, the same pattern, the same composition, the same things present, or the same mix of measurements and categories. The flowchart and the table below summarize the choices.

| `distance_method` | Similar means… | Typical data | Scale first? |
|---|---|---|---|
| `"euclidean"` | close values in every variable, with large single differences counting heavily | measurements in different units (pH, mg/L, temperature) | yes, when units differ |
| `"manhattan"` | close values in every variable, with one extreme difference counting less | as euclidean, when outliers or spikes are a concern | yes, when units differ |
| `"correlation"` | the same pattern across the variables, regardless of size | profiles at different concentrations or totals | optional |
| `"cosine"` | pointing in the same direction, regardless of length | embeddings and other high-dimensional data | optional |
| `"bray_curtis"` | sharing the same amounts of the same components | abundances and percentages, especially with many zeros | no |
| `"aitchison"` | the same proportions between components | compositions summing to 100% with few zeros | no |
| `"jaccard"` | containing the same things, amounts ignored | presence/absence, detected or not | no |
| `"gower"` | matching across a mix of numbers and categories | tables with categorical columns | handled internally |